

BALWA MOHAMMADALI

Machine Learning Engineer

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EDUCATION

Bachelor of Engineering (B.E.) in Computer Engineering

2023 – Present

Gujarat Technological University, Ahmedabad

CGPA: 9.26 / 10

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Scikit-Learn, Django, FastAPI, Pydantic, Tensorflow/keras

ML/DL: Supervised Learning, Unsupervised Learning, NLP, ANN, CNN, RNN, LSTM

Tools & Platforms: Git/GitHub, VS Code, Jupyter Notebook, Google Colab, MySQL, Excel, n8n

EXPERIENCE

Generative AI Intern – SmartBridge | Remote

3 Jul 2026 – 17 Jul 2026

- Completed NASSCOM FutureSkills Prime program covering GenAI, LLMs, prompt engineering, and AI agents.
- Built a Generative AI web application using Antigravity IDE with AI-assisted development workflows.

Machine Learning Intern – Edunet Foundation | Remote

13 May 2025 – 13 Jun 2025

- Developed an end-to-end Heart Disease Prediction system using ML, including preprocessing, feature engineering, and model evaluation.
- Performed EDA and visualization using Pandas/NumPy and trained classification models with Scikit-Learn.

PROJECTS

Plant Disease Detection System | EfficientNetB0, TensorFlow/Keras, FastAPI, Grad-CAM

- Built a deep learning-based plant disease classification system using EfficientNetB0 transfer learning to classify images into 39 plant health and disease categories.
- Applied image preprocessing and augmentation and fine-tuned the final 20 layers of EfficientNetB0, achieving 99.15% validation accuracy.
- Evaluated model performance using precision, recall, and F1-score, achieving ~99% macro precision, recall, and F1-score across 11,105 validation samples.
- Implemented Grad-CAM for visual model explainability and integrated the trained model with a FastAPI prediction API for image-based inference.

Movie Recommendation System | NLP, Scikit-Learn, FastAPI

- Built a content-based movie recommendation system using movie metadata and NLP techniques.
- Performed data preprocessing and feature engineering by combining genres, keywords, cast, crew, and movie overview into a unified feature set.
- Applied CountVectorizer to convert textual features into numerical vectors and used cosine similarity to generate movie recommendations.
- Integrated TMDB API to fetch movie posters and display recommendations through an interactive web interface.

HACKATHONS & COMPETITIONS

- **Smart India Hackathon 2025** – Led the team in developing a CNN-based cattle and buffalo breed recognition system using MobileNet, TensorFlow/Keras, and Django, achieving ~74% validation accuracy with real-time prediction and Top-3 results.

- **Odoo Hackathon 2025** – Designed and developed a sustainable clothing exchange platform with point-based redemption and inventory management workflows.
- **UNESCO Global Youth Hackathon 2025** – Built an NLP and machine learning-based fake news detection system for real-time text classification and prediction.

CERTIFICATIONS

- Python – HackerRank
- Machine Learning – IBM SkillsBuild
- Java Programming – Great Learning